

**KROONSTAD, South Africa**WMO: **683550**Lat: **27.666S**Lon: **27.314E**Elev: **4705**StdP: **12.36**Time Zone: **2.00 (E02)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 32.0       | 34.6 | 6.2                           | 9.0 | 61.6 | 9.5 | 10.6 | 56.8 | 24.8                  | 63.3 | 21.0 | 57.7 | 5.8                   | 100  | 0.442 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 20.3                   | 91.6            | 59.5 | 89.1 | 59.3 | 86.9 | 59.3 | 68.2                | 76.8 | 66.1 | 77.0 | 65.1 | 76.0 | 10.8                 | 270  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 65.8                            | 113.4 | 71.2 | 63.5 | 104.5 | 69.1 | 62.0 | 99.2 | 68.8 | 35.5          | 75.3 | 33.6 | 76.9 | 32.8 | 75.8 | 75.0              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 22.2          | 19.7 | 17.5 | DB                         | 27.8 | 96.7               | 3.7 | 3.8                                                | 25.1 | 99.5       | 23.0 | 101.7      | 20.9 | 103.9      | 18.3 | 106.6 | (4) |
| (5)               |      |      | WB                         | 24.9 | 68.4               | 2.6 | 1.8                                                | 23.0 | 69.8       | 21.4 | 70.8       | 20.0 | 71.9       | 18.1 | 73.2  | (5) |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 64.3   | 73.6  | 71.7  | 69.5  | 64.2  | 57.7  | 51.9  | 51.2  | 56.8  | 63.9  | 68.6  | 70.1  | 72.8  |  |  |
|                                                                        | DBStd       | 9.41   | 4.42  | 4.09  | 5.14  | 5.14  | 5.66  | 4.81  | 4.91  | 6.38  | 6.56  | 6.06  | 5.50  | 4.80  |  |  |
|                                                                        | HDD50       | 115    | 0     | 0     | 0     | 1     | 14    | 34    | 47    | 17    | 2     | 0     | 0     | 0     |  |  |
|                                                                        | HDD65       | 1566   | 2     | 5     | 22    | 73    | 230   | 392   | 428   | 261   | 97    | 35    | 17    | 4     |  |  |
|                                                                        | CDD50       | 5335   | 733   | 607   | 606   | 427   | 252   | 92    | 84    | 229   | 420   | 577   | 602   | 706   |  |  |
|                                                                        | CDD65       | 1310   | 270   | 192   | 162   | 49    | 3     | 0     | 0     | 8     | 65    | 146   | 170   | 245   |  |  |
|                                                                        | CDH74       | 10411  | 2141  | 1229  | 1038  | 290   | 55    | 2     | 1     | 106   | 719   | 1353  | 1468  | 2009  |  |  |
|                                                                        | CDH80       | 3397   | 789   | 345   | 270   | 43    | 2     | 0     | 0     | 7     | 201   | 468   | 513   | 759   |  |  |
| (14) Wind                                                              | WSAvg       | 8.3    | 9.3   | 8.3   | 7.7   | 6.7   | 6.2   | 6.7   | 6.9   | 8.3   | 9.2   | 9.9   | 10.4  | 10.2  |  |  |
| (15) Precipitation                                                     | PrecAvg     | 23.3   | 3.6   | 3.1   | 3.3   | 2.2   | 0.8   | 0.3   | 0.2   | 0.4   | 0.8   | 2.4   | 3.1   | 3.0   |  |  |
|                                                                        | PrecMax     | 34.4   | 8.6   | 7.9   | 8.7   | 6.8   | 4.3   | 1.9   | 1.7   | 5.0   | 6.7   | 8.9   | 9.6   | 8.8   |  |  |
|                                                                        | PrecMin     | 11.6   | 0.7   | 0.2   | 0.3   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.6   | 0.3   |  |  |
|                                                                        | PrecStd     | 4.9    | 2.1   | 1.8   | 1.9   | 1.9   | 1.0   | 0.4   | 0.4   | 0.9   | 1.2   | 1.9   | 1.9   | 1.6   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 95.2  | 90.8  | 90.0  | 84.9  | 78.7  | 72.0  | 72.9  | 80.7  | 88.7  | 91.8  | 92.9  | 96.0  |  |  |
|                                                                        |             | MCWB   | 61.1  | 60.9  | 61.0  | 58.8  | 53.8  | 49.4  | 49.0  | 53.0  | 55.1  | 57.1  | 57.7  | 59.7  |  |  |
|                                                                        | 2%          | DB     | 90.6  | 87.4  | 86.1  | 80.2  | 75.2  | 68.9  | 69.5  | 76.9  | 85.5  | 88.5  | 89.2  | 90.9  |  |  |
|                                                                        |             | MCWB   | 61.1  | 61.0  | 60.8  | 58.0  | 52.9  | 48.5  | 47.6  | 50.9  | 54.5  | 56.3  | 57.9  | 60.2  |  |  |
|                                                                        | 5%          | DB     | 87.6  | 84.3  | 82.9  | 77.2  | 72.5  | 66.1  | 66.7  | 73.5  | 82.0  | 85.3  | 85.8  | 87.3  |  |  |
|                                                                        |             | MCWB   | 62.2  | 62.0  | 60.9  | 57.3  | 52.4  | 47.7  | 46.6  | 49.7  | 53.1  | 55.6  | 57.8  | 60.7  |  |  |
|                                                                        | 10%         | DB     | 84.5  | 81.6  | 80.0  | 74.4  | 69.5  | 63.4  | 63.5  | 69.9  | 78.2  | 81.8  | 82.4  | 84.0  |  |  |
|                                                                        |             | MCWB   | 62.0  | 62.3  | 60.6  | 56.9  | 51.7  | 47.0  | 45.5  | 48.2  | 51.8  | 55.3  | 57.6  | 60.9  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 69.8  | 67.2  | 66.1  | 63.3  | 64.6  | 59.3  | 51.2  | 55.0  | 59.1  | 63.0  | 64.5  | 72.3  |  |  |
|                                                                        |             | MCDB   | 81.7  | 77.1  | 76.2  | 74.5  | 68.6  | 62.3  | 66.4  | 74.6  | 78.0  | 76.2  | 76.4  | 76.5  |  |  |
|                                                                        | 2%          | WB     | 67.0  | 66.2  | 64.8  | 61.8  | 58.2  | 53.3  | 49.4  | 53.1  | 57.0  | 61.0  | 63.0  | 70.1  |  |  |
|                                                                        |             | MCDB   | 76.7  | 77.1  | 75.8  | 71.9  | 65.2  | 58.9  | 65.0  | 71.4  | 75.5  | 75.0  | 75.2  | 75.5  |  |  |
|                                                                        | 5%          | WB     | 65.7  | 65.4  | 63.9  | 60.5  | 55.3  | 50.8  | 47.8  | 51.6  | 55.4  | 59.7  | 62.0  | 66.5  |  |  |
|                                                                        |             | MCDB   | 76.1  | 76.0  | 75.1  | 70.1  | 65.7  | 61.1  | 63.3  | 69.1  | 74.8  | 73.9  | 74.4  | 78.7  |  |  |
|                                                                        | 10%         | WB     | 64.8  | 64.6  | 62.9  | 59.1  | 53.5  | 48.6  | 46.5  | 49.8  | 54.0  | 58.5  | 61.1  | 64.1  |  |  |
|                                                                        |             | MCDB   | 75.4  | 74.9  | 73.9  | 69.7  | 64.9  | 60.2  | 61.6  | 66.6  | 72.6  | 72.9  | 73.5  | 76.8  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 20.3  | 18.9  | 19.4  | 20.7  | 24.3  | 24.6  | 25.7  | 25.6  | 26.7  | 24.4  | 23.1  | 21.1  |  |  |
|                                                                        |             | MCDBR  | 25.5  | 23.8  | 24.0  | 24.0  | 27.2  | 27.3  | 28.4  | 28.6  | 29.7  | 28.4  | 27.5  | 26.5  |  |  |
|                                                                        |             | MCWBR  | 6.7   | 6.9   | 6.8   | 7.7   | 10.8  | 12.7  | 12.8  | 12.0  | 11.0  | 9.0   | 8.1   | 7.3   |  |  |
|                                                                        |             | MCDBR  | 17.5  | 17.7  | 19.1  | 18.5  | 22.0  | 22.5  | 25.9  | 25.6  | 26.5  | 22.8  | 21.3  | 19.1  |  |  |
|                                                                        | 5% WB       | MCWBR  | 6.0   | 5.7   | 6.8   | 6.8   | 10.2  | 12.5  | 11.9  | 10.8  | 10.2  | 7.8   | 7.2   | 6.8   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.368 | 0.369 | 0.355 | 0.314 | 0.283 | 0.268 | 0.267 | 0.322 | 0.362 | 0.388 | 0.344 | 0.353 |  |  |
|                                                                        | taud        |        | 2.359 | 2.359 | 2.370 | 2.461 | 2.525 | 2.532 | 2.512 | 2.350 | 2.241 | 2.211 | 2.382 | 2.391 |  |  |
|                                                                        | Ebn at Noon |        | 309   | 303   | 298   | 296   | 292   | 291   | 295   | 288   | 290   | 294   | 314   | 314   |  |  |
|                                                                        | Edh at Noon |        | 42    | 41    | 38    | 32    | 27    | 25    | 27    | 35    | 43    | 47    | 41    | 41    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2245  | 2073  | 1825  | 1531  | 1334  | 1202  | 1313  | 1568  | 1907  | 2087  | 2274  | 2308  |  |  |
|                                                                        | RadStd      |        | 212   | 186   | 134   | 116   | 60    | 56    | 41    | 78    | 85    | 140   | 154   | 133   |  |  |

**Historical Trends**

|                             |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|-----------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                             |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) Regional (8 neighbors) |  | +1.19 | +1.35   | N/A    | N/A   | N/A     | N/A   | N/A | -326        | +421  | +157  |       |

Nomenclature: See separate page